

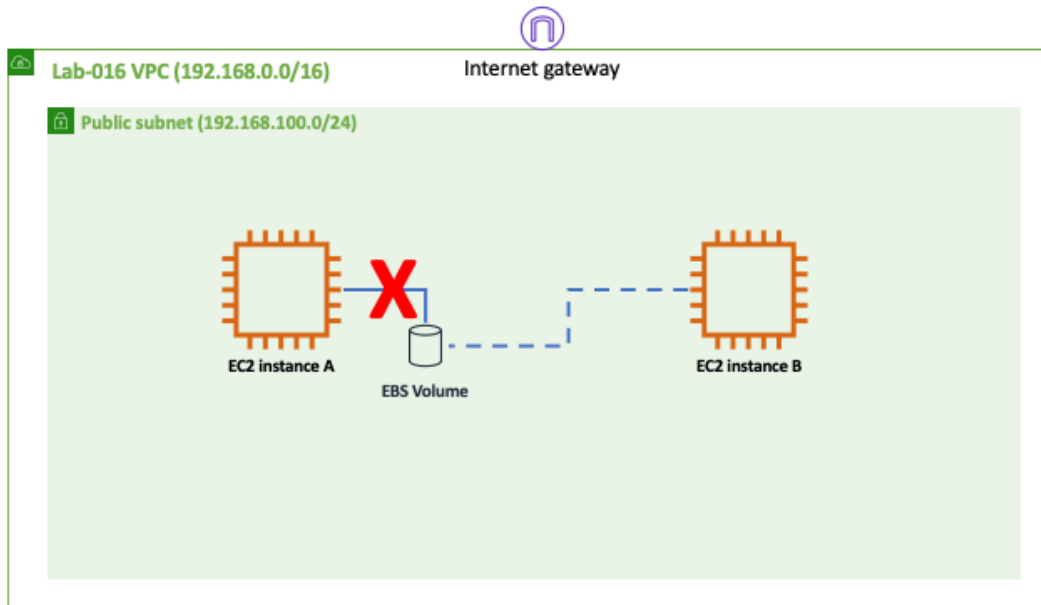
Lab-016

Moving an EBS Volume Between EC2 Instances

Goal

The goal of this lab is to illustrate how to move an EBS volume from one EC2 instance to another.

Architecture Diagram



Overview

Step 1 - Create an EBS Volume

During the *Add Storage* step of *Launch Instance*, add a new EBS volume.

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type ⁽ⁱ⁾	Device ⁽ⁱ⁾	Snapshot ⁽ⁱ⁾	Size (GiB) ⁽ⁱ⁾	Volume Type ⁽ⁱ⁾	IOPS ⁽ⁱ⁾	Throughput (MB/s) ⁽ⁱ⁾	Delete on Termination ⁽ⁱ⁾	Encryption ⁽ⁱ⁾
Root	/dev/xvda	snap-05dbd93527b8f6f59	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒	Not Encrypted

Add New Volume

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage. [Learn more](#) about free usage tier eligibility and usage restrictions.

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type ⁽ⁱ⁾	Device ⁽ⁱ⁾	Snapshot ⁽ⁱ⁾	Size (GiB) ⁽ⁱ⁾	Volume Type ⁽ⁱ⁾	IOPS ⁽ⁱ⁾	Throughput (MB/s) ⁽ⁱ⁾	Delete on Termination ⁽ⁱ⁾	Encryption ⁽ⁱ⁾
Root	/dev/xvda	snap-05dbd93527b8f6f59	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒	Not Encrypted
EBS	/dev/sdb	Search (case-insensit)	8	General Purpose SSD (gp2)	100 / 3000	N/A	☐	Not Encrypted

Add New Volume

Step 2 - Mount EBS Volume

Connect to the EC2 instance via ssh and run `lsblk` to learn your EBS device's name. For example, the output I got from `lsblk` was:

NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT

```
NAME      MAJ:MIN RM  SIZE RO  TYPE MOUNTPOINT
xvda      202:0    0   8G  0 disk
└─xvda1   202:1    0   8G  0 part /
xvdb      202:16   0   8G  0 disk
```

The device `/dev/xvda` is the boot volume, mounted at the root level. The device `/dev/xvdb` must be the EBS volume that we created. Run the following commands to format and mount the disk:

```
sudo mkfs.ext4 /dev/xvdb
mkdir data
sudo mount /dev/xvdb data
```

Then use the following commands to create some files in your new disk:

```
sudo touch data/test1.txt
sudo touch data/test2.txt
sudo touch data/test3.txt
```

Step 3 - Terminate the EC2 Instance

This step illustrates the fact that when you terminate an EC2 instance, non-boot volumes are not deleted by default. Go to *Elastic Block Store - Volumes* and verify that your EBS volume is still available even after the EC2 instance is terminated.

Create VolumeActions

Filter by tags and attributes or search by keyword

☐	Name	Volume ID	Size	Volume Type	IOPS	Snapshot	Created
☐		vol-068d61b2538c69329	8 GiB	gp2	100		June 26, 2020 at 3:...

Step 4 - Attach EBS Volume

Launch a new EC2 instance but this time do not create a new EBS volume. Instead, after your instance is launched, attach the EBS volume created earlier to the EC2 instance.

Create Volume

Actions ^

Filter by tags

☐	Name	Size	Volume Type	IOPS
☐		8 GiB	gp2	100
☒		8 GiB	gp2	100

Modify Volume

Create Snapshot

Delete Volume

Attach Volume

Detach Volume

Force Detach Volume

Change Auto-Enable IO Setting

Add/Edit Tags

Attach Volume

×

Volume ⓘ

vol-068d61b2538c69329 in us-west-1a

Instance ⓘ

i-0a8f7aa2a75e39484

in us-west-1a

Device ⓘ

/dev/sdf

Linux Devices: /dev/sdf through /dev/sdp

Note: Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

Cancel

Attach

Test and Validation

Connect to the new EC2 instance, mount the EBS volume and see if you can access the files created earlier.